TRACE-RX-M

Adapted Reference-Comparison with Measured-Survival Reliability

for Robust AI-Generated Image Detection

Architecture and method report · Version 2 · TikTok TechJam 2026 · 31 August 2026

Status. A falsifiable design, not a performance claim. No benchmark result is claimed here. **Change from v1:** the encoder is now LoRA-adapted rather than frozen, and detection is framed as reference-comparison against an authentic memory. This is a deliberate architectural commitment with consequences throughout, most importantly the promotion of data alignment and generator diversity from advisable to mandatory (Sec. 7). There is no ablation budget; every component is therefore justified by a stated mechanism, and Section 11 defines the single measurement that validates or refutes the whole approach.

1 Thesis and design constraint

Two commitments define this system.

Reference-comparison over artifact classification. Detection is framed as verifying consistency against a learned authentic reference, not as fitting the distribution of generator artifacts. The artifact distribution is non-stationary; the authentic distribution is not.

Measured survival over test-time intervention. Whether an expert's evidence is still present at a degraded endpoint is estimated once, as a population quantity indexed by observable image quality, rather than probed per-image by perturbation. The latter is not identifiable from a single endpoint; the former is.

1.1 The constraint that shapes component selection

There is no budget for architecture ablations. This changes the selection rule. It is not sufficient that a component performed well in a published system: it must be justified by a mechanism that applies to *this* configuration, and no two components may rest on contradictory assumptions. A conflict discovered mid-run cannot be resolved.

The governing variable is **how much the encoder is permitted to move**, and the source literature disagrees about it:

Approach	Encoder	Where its gain comes from
Frozen probing	Frozen	Precisely from *not* adapting: fixed features cannot absorb generator-specific artifacts, and the constraint is what protects generalisation.
Reference-comparison memory	LoRA-adapted	From adapting features so that the residual against a fixed authentic memory becomes discriminative.
Data alignment	Agnostic	Premised on the fact that a detector *with* capacity will find shortcuts unless they are removed from the data.

This design takes the adapted branch, which is a bet: adaptation buys the structured reference-comparison objective at the cost of the protection that freezing provides. Section 7 records what that makes mandatory, and Section 11 records how the bet is checked.

2 Architecture

2.1 Overview

image
$\rightarrow$ LoRA-adapted DINOv3-L $\rightarrow$ patch tokens F
$\rightarrow$ frozen real-only memory $\mathbf{M}$, sparse top-k retrieval $\rightarrow$ reference $\hat{F}$
$\rightarrow$ residual $\Delta F = F - \hat{F}$ + retrieval statistics $(s_{\max}, s_{\text{ent}})$
$\rightarrow$ evidence head $\rightarrow$ logit z
$\rightarrow$ availability $a(x)$ from quality cell $\rightarrow$ temperature $\rightarrow$ pred

2.2 Backbone

The backbone is selected on dense patch-feature quality, since the memory reconstructs patch tokens and the residual is only as meaningful as the space it is computed in. DINOv3-L is the operating choice.

Licence blocker. DINOv3's licence is gated and must be cleared against competition eligibility rules *before* the build begins. This is administrative rather than technical, but it is a hard blocker. If it fails, substitute the strongest licence-cleared dense encoder available and re-run the capacity fit of Section 2.4; nothing else in the design changes.

2.3 Phase 1: authentic memory

A memory bank $\mathbf{M} \in \mathbb{R}^{K \times D}$ of authentic prototypes is fitted on real images only. Patch features are projected onto the span of $\mathbf{M}$ by sparse cross-attention:

$$\mathbf{A} = \text{Softmax}\left(\text{Top-}k\left(\frac{QK^\top}{\sqrt{D}}\right)\right), \qquad \hat{F} = \mathbf{A}V,$$

with Q from F and K, V from $\mathbf{M}$, trained on authentic data alone under

$$\mathcal{L}_{\text{Phase 1}} = \|F - \hat{F}\|_2^2 + \lambda \, \|\mathbf{M}\mathbf{M}^\top - \mathbf{I}\|_F.$$

On the orthogonality term. When $K > D$, exact orthogonality is unachievable: K vectors in D dimensions cannot be mutually orthogonal, so $\|\mathbf{M}\mathbf{M}^\top - \mathbf{I}\|_F$ has a nonzero lower bound. It is a *diversity pressure* discouraging redundant prototypes, not a constraint that can be satisfied. Report it as such.

2.4 Capacity fitting: K and top-k are derived, not inherited

Published memory configurations were fitted on corpora roughly five times the size of ours. Capacity does not transfer across dataset scale, and the failure is asymmetric: an undersized dictionary fails to cover the authentic manifold, so "unexplainable" comes to mean *semantically unusual* rather than *synthetic*. Rare-but-authentic content — scans, screenshots, documents, CGI, unusual composition — then produces large residuals and is flagged as generated. This is a false-positive mechanism on exactly the authentic long tail the design is obliged to protect.

K and top-k are therefore fixed by a *coverage criterion* on cached tokens before Phase 2 begins. This is a capacity fit, not an architecture ablation:

1. Fit prototypes on the authentic training masters.
2. On a held-out, **subtype-balanced** authentic set, measure mean reconstruction error and the fraction of patches exceeding a tail threshold.
3. Increase K until the *authentic tail* stops improving, not until mean error stops improving. The tail is what governs false positives.
4. Select top-k by the same criterion. Too small underfits complex texture; too large lets generative structure be absorbed into the reconstruction, narrowing the residual gap.

Record the chosen values and the coverage curve. If the curve is still rising at the largest K tested, say so — that is an admission of undersizing, and it predicts the direction of failure.

2.5 Phase 2: adapted reference-comparison

The memory is frozen. The encoder is LoRA-adapted. Evidence is drawn from two complementary signals: *how confidently* the reference was retrieved, and *what remained unexplained*.

$$V_{\mathrm{per}} = \mathrm{MLP}\big([s_{\max}, s_{\mathrm{ent}}]\big), \qquad V_{\mathrm{res}} = \mathrm{Linear}\big(\mathrm{pool}(\Delta F)\big),$$

$$z = \mathrm{MLP}_{\mathscr{C}}\big(\mathrm{Concat}[V_{\mathrm{per}}, V_{\mathrm{res}}]\big).$$

Deliberate deviation: pool the residual as a vector, not a scalar. The motivating claim for a memory is that *some residual directions* indicate rare-but-authentic content while others accompany generation. Pooling ΔF to scalar magnitudes destroys direction before the head sees it, leaving only "how far from the dictionary," and the architecture can then no longer do what the intuition promises. $\mathrm{pool}(\Delta F)$ is therefore mean- and tail-pooled over patches into a D-dimensional vector. Retrieval statistics remain scalar.

2.6 What the memory contributes under adaptation

With a frozen encoder, a real-only memory is a genuinely stable anchor: features are fixed, the dictionary is fixed, and labels reach only a small head. Under LoRA this is no longer true in the strong form. The encoder can move features so that authentic images reconstruct well and generated images reconstruct badly, rather than discovering that they already do. The residual gap becomes partly manufactured by training.

The memory remains valuable — it is what gives adaptation a structured objective, and without it LoRA on a binary target is ordinary classifier fine-tuning with the full artifact-overfitting failure mode. But the claim must be stated accurately.

Claim discipline. Do not write that the memory is an anchor immune to generator drift. Under LoRA it is a *structured inductive bias*, not an invariant. The defensible statement is that adaptation is constrained to act through reconstruction against an authentic dictionary rather than through free discrimination.

3 Degradation model

3.1 The lattice hypothesis

Sorting the transform suite by energy removed does not predict difficulty. Blur at $\sigma = 2$ removes more high-frequency content in several bands than JPEG at $q = 30$, yet is more survivable. Sorting by effect on the sampling lattice does predict it.

Class	Operators	Effect on the pixel lattice
Preserving	blur $\sigma \le 2$; crop 80%; colour $\pm 20\%$	Convolution on the same grid; a subset of the same grid at native resolution; a per-pixel value remap. Neighbour relations survive.
Destroying	resize $0.25\times$; JPEG $q30$; noise $\sigma = 0.10$	Resampling onto a new grid; block-wise coefficient quantisation with an imposed 8×8 structure; broadband additive energy on every sample.

H1. Dominant forensic evidence is lattice-dependent — upsampling signature, interpolation trace, short-range neighbour relations — rather than raw high-frequency energy.

3.2 Three mechanisms

Erasure (resize). A deterministic many-to-one map. By the data processing inequality, $I(T(X);Y) \le I(X;Y)$, and for $0.25\times$ decimation every component above the new Nyquist limit is gone. No aggregation, augmentation or fusion recovers it. This is a floor, not a defect.

Quantisation (JPEG). Each DCT coefficient is mapped $c \mapsto \mathrm{round}(c/q) \cdot q$. Evidence in a coefficient of amplitude below $q/2$ is annihilated; above $q/2$ it largely survives. The response is a *threshold*, which is why repeated application at the same quality saturates rather than degrading proportionally. JPEG also *adds* an 8×8 grid, making it simultaneously destructive and a potential class shortcut whenever authentic and synthetic pools differ in codec history.

Masking (noise). Additive, and not information-destroying in the same sense. Evidence of per-pixel amplitude a sits at $\mathrm{SNR} = a^2/\sigma^2$; for residual traces with $a \approx 0.01$–0.02 and $\sigma = 0.10$ this is roughly -14 to -20 dB. The signal is present but buried.

H2. Under masking, evidence is recoverable by aggregation: for spatially coherent evidence and approximately white noise, averaging n patch responses gives SNR gain $\propto n$. Under erasure and quantisation it is not, because the operator acts identically everywhere and the loss is not sample-wise independent. Patch aggregation should therefore retain mean, standard deviation and upper quantile so the downstream layer can express this regime dependence rather than committing to one statistic.

4 The false-availability trap

Erasure operators *lower* high-frequency energy. Masking *raises* it. Any scalar high-frequency-rolloff descriptor is therefore non-monotone in evidence quality: a noise-destroyed image presents the same high-frequency statistics as a clean, highly detailed one.

Two failures follow, and the second is severe:

1. **Mis-binned normalisation.** A noisy image assigned to a clean quality cell is normalised against the wrong authentic reference distribution.
2. **False availability.** In reliability-weighted fusion, high measured high-frequency energy reads as *signal present, trust this expert*. Maximum trust is assigned to an expert whose input is buried in noise. This is worse than having no reliability layer at all.

4.1 Noise-floor-subtracted structural energy

Estimate the noise floor independently (wavelet-MAD gives a robust $\hat{\sigma}$) and remove its expected contribution before using high-frequency energy as a quality descriptor:

$$E_{\mathrm{HF}}^{\mathrm{struct}} = \max\left(0,\ E_{\mathrm{HF}} - \hat{\sigma}^2 \cdot \frac{|B_{\mathrm{HF}}|}{|B_{\mathrm{total}}|}\right).$$

The quality vector uses $E_{\mathrm{HF}}^{\mathrm{struct}}$, never raw E_{HF}.

Required audit. Subtraction is necessary but not sufficient. Confirm from the occupancy table that noise-degraded images occupy *distinct* cells rather than collapsing into clean ones. With few cells there may be insufficient resolution to separate them.

5 Measured-survival availability

5.1 Why not interventions

A controlled intervention estimates $|z(x) - z(T_p(x))|$ and reads a small response as robustness. This is not identifiable. Both "evidence robust to T_p" and "evidence already destroyed before observation" produce a small response, and no function of a single endpoint separates them; a clean counterpart does not exist for a redistributed image. Interventions also multiply inference by the number of operators, and publishing the operator set creates an attack surface — an adversary who knows it can optimise an image to be flat under exactly those perturbations.

5.2 The identifiable quantity

Availability is not estimable per image. It *is* estimable per cell. With $b(x)$ a quality cell assigned from observable endpoint properties only:

$$d'(k,b) = \frac{\mu_{\mathrm{AI}}(k,b) - \mu_{\mathrm{real}}(k,b)}{\sqrt{[\sigma^2_{\mathrm{AI}}(k,b) + \sigma^2_{\mathrm{real}}(k,b)]/2}}, \qquad a_k(x) = \mathrm{clip}\left(\frac{d'(k,b(x))}{d'(k,\mathrm{clean})}, 0, 1\right).$$

This is a lookup table fitted to *measured discriminability*, not a gate trained through the fusion loss. It is identifiable, because $d'(k,b)$ is a population quantity with a consistent estimator. It is saturation-free, because an already-degraded image lands in a low-quality cell regardless of processing history — the ambiguity that defeats interventions never arises. And it is free at inference, requiring no additional forward passes.

Cells are formed by binning

$$q(x) = \left[\log\min(H,W),\ \hat{\sigma},\ q_{\mathrm{block}},\ E^{\mathrm{struct}}_{\mathrm{HF}}\right],$$

with per-cell statistics fitted on a dedicated authentic-null subset, lineage-disjoint from every other role, small cells shrunk toward global authentic statistics.

5.3 Fitting order under adaptation

The table is fitted *after* Phase 2, on held-out data, using the adapted model. This is correct — survival should be measured for the model that ships. It does mean the table is more specific to the observed transform suite than it would be for a frozen model, which raises the stakes on the held-out-family check in Section 11.

5.4 Output

With availability weighting applied to the single primary expert:

$$z_{\mathrm{fused}}(x) = a(x)\,\tilde{z}(x), \qquad \mathrm{pred}(x) = \sigma\left(\frac{z_{\mathrm{fused}}(x)}{T}\right),$$

T fitted on an independent calibration split after all architecture choices freeze. If the optional native forensic branch is admitted, combine by L_2-regularised logistic stacking over the two availability-weighted logits, fitted out-of-fold.

The inference contract returns a numeric score for every image. “Insufficient evidence” is a reporting overlay driven by the risk–coverage curve, never a missing prediction or a third class.

6 Data

6.1 Composition

Class	Share	Role
Authentic	50%	Memory fitting (Phase 1), supervised negatives, quality-cell null subset. Subtype-balanced.
Native text-to-image	$\approx 37.5\%$	The primary positive class. Must span diffusion, DiT, autoregressive and relevant GAN families.
DDA-aligned	$\approx 12.5\%$	Auxiliary hard near-real counterfactuals. Not native members of the target class.

Why DDA is a minority, not half. A DDA sample is a VAE reconstruction of a specific real image, frequency-matched and pixel-mixed with it — by construction nearly identical to its source. It is a training device, not a purely generated image. At parity within the positive class, an adapted encoder is asked to find what separates $\{\text{text-to-image}\} \cup \{\text{VAE-reconstructed real}\}$ from real, and may settle on a compromise rule that serves neither. Published DDA results were obtained with *no* text-to-image images in training, so the mixture ratio is unstudied. Keep DDA at 10–20% of positives and admit it through the paired ranking auxiliary

$$\mathscr{L}_{\text{pair}} = \text{softplus}\big(m - z(x_{\text{DDA}}) + z(x_{\text{real-source}})\big),$$

rather than at equal weight in the primary objective. It should shape the boundary, not define the class.

6.2 Generator diversity is the binding constraint

Volume is not the limiting factor; a LoRA adapter has few trainable parameters and published systems in this family train on far less. The limiting factor is **how many independent things the label correlates with**.

Frozen-encoder methods have generalised from a single training generator precisely because freezing constrains what can be memorised. Adaptation removes that constraint. Under LoRA, generator diversity stops being advisable and becomes structural: with only a few generator families, the adapter will fit those decoders and collapse on unseen ones.

Reportable number. State the generator family count explicitly. If it is below roughly eight, that is the highest-priority fix in the entire build — higher than any architectural choice in this document.

6.3 Transformations

Generate transformations **on the fly**, not as a pre-materialised set. A pre-materialised one-endpoint-per-master corpus is a fixed, sampled-once augmentation: it doubles storage and adds no variety. Sampling per epoch gives a different degradation view of each master on every pass, at zero storage cost. Evaluation transforms remain deterministic with a pinned resampling kernel.

Apply all transformations **symmetrically by label**, including the three hard operators. If the DDA and text-to-image subsets receive different transform distributions, or if authentic and synthetic pools are transformed at different rates, the transform itself becomes a class cue — an exploit a frozen probe would struggle to use and an adapted encoder will find easily.

6.4 Partition integrity

A master, its prompt siblings, its duplicate components, **its DDA derivatives**, and every transformed view remain in one partition. DDA pairs deserve explicit naming in the manifest: a DDA sample and its source real image are near-identical by construction, so splitting them across train and evaluation is the most damaging form of near-duplicate leakage available.

7 What adaptation makes mandatory

Three items that were advisable under a frozen encoder are load-bearing under LoRA.

1. **Format and frequency alignment.** A frozen encoder with a small head has limited capacity to exploit codec, resolution and spectral shortcuts. Adaptation restores that capacity. Canonicalise or matched-re-encode classes, and note that published reference-comparison work JPEG-compresses its training positives specifically to match the real images' format.

2. **The nuisance probe runs twice.** Before training, verify that metadata-, codec- and size-only baselines are near chance. **After** training, re-run the probe against the adapted model's predictions. The second check is the one that matters, and it is the cheapest available insurance against shipping a codec detector.

3. **Symmetric hard-transform augmentation with a held-out family.** An adapted encoder can partially learn to compensate for degradation it has seen and will not extend that to degradation it has not.

8 Training methodology

The build proceeds in six stages. Each stage names what is trainable, what data it may see, and the condition under which the next stage is permitted to begin. Stages are strictly ordered: a later stage never modifies an artefact frozen by an earlier one, and no stage may consume data reserved for a later role.

8.1 Trainable-parameter ledger

Stage	Trainable	Frozen	Data permitted
S0 Protocol	—	—	Manifests, probes only
S1 Cache	—	Encoder	All permitted lineages
S2 Memory	**M**	Encoder	Authentic memory pool only
S3 Capacity	—	**M** candidates	Authentic capacity-validation pool
S4 Detection	LoRA + heads	**M**	Supervised pool (real, T2I, DDA)
S5 Reliability	—	Everything above	Authentic null + development
S6 Calibration	T, thresholds	Everything above	Calibration pool only

The one-way rule. The memory is fitted in the *un-adapted* encoder's feature space and then frozen while the encoder moves beneath it. This is intentional — it is what gives adaptation a fixed authentic target to optimise against — but it means Stage 2 must never be re-run after Stage 4 begins. Refitting the memory against adapted features would let the reference chase the representation, and the residual would lose its meaning entirely.

8.2 S0 — Protocol and confound gate

Before any training. Verify licences, lineage integrity (masters, prompt siblings, duplicate components, DDA pairs and transformed views each confined to one partition), and resampler parity between training augmentation and evaluation. Train metadata-, codec- and dimension-only baselines and confirm they sit near chance. Document the patch-extractor's behaviour at the lowest evaluation resolution.

Gate: nuisance AUC at or near chance, zero leakage. Fix data before touching architecture.

8.3 S1 — Feature caching

Extract patch tokens for the authentic memory and capacity pools using the *un-adapted* encoder. These tokens serve Stages 2 and 3 only; Stage 4 recomputes features every step because the encoder is being adapted. Exclude special and register tokens from reconstruction.

Gate: cached token statistics match a direct forward pass on a held-out sample.

8.4 S2 — Memory fitting

Fit **M** on authentic patch tokens alone under $\mathscr{L}_{\text{Phase 1}}$. Initialise prototypes from k-means on normalised tokens to avoid dead entries, then optimise. Normalise both tokens and prototypes; compute attention logits and entropy in FP32 even where cached tokens are BF16; score prototypes in chunks and gather only selected values.

Monitor prototype usage throughout. A prototype that is never selected is wasted capacity; a prototype that monopolises retrieval indicates collapse. Report the usage histogram.

Gate: no dead or monopolising prototypes; held-out authentic reconstruction error stable across the authentic subtypes, not merely low on average.

8.5 S3 — Capacity selection

Run the coverage criterion of Section 2.4 over candidate $(K, \text{top-}k)$ on the subtype-balanced authentic validation pool. Select on the *authentic tail*, not mean reconstruction error. This consumes no labelled synthetic data and is not an architecture ablation.

Gate: coverage curve flattened, or undersizing explicitly recorded with its predicted failure direction.

8.6 S4 — Detection training

The only stage with a substantial compute cost. The memory is frozen; LoRA adapters on the encoder, the perplexity MLP, the residual projection and the classifier head are trainable.

- **Optimiser.** AdamW, cosine schedule with short warm-up, BF16. Separate parameter groups for LoRA and heads; heads tolerate a higher learning rate than adapters.
- **Augmentation.** Sampled on the fly, symmetrically by label, covering the full transform suite with the three hard operators represented at their evaluation severities. One held-out transform family is excluded entirely.
- **Batch composition.** Balanced by class, and within the positive class by generator family, so that no single generator dominates the gradient. DDA samples occupy their reserved minority share and contribute through $\mathcal{L}_{\text{pair}}$.
- **Monitoring.** Track authentic-subtype loss separately from the aggregate. A rising worst-subtype loss against a falling mean is the signature of the memory-undersizing failure and should stop the run.

Adaptation strength is a risk dial, not a hyperparameter to maximise. LoRA rank and learning rate control how far the representation may drift from its pretrained state. Larger values fit the training generators better and generalise to unseen ones worse — the exact trade this design is betting on. Start conservative (low rank, adapters at a fraction of the head learning rate). Since there is no budget to sweep this, the held-out-family check of Section 11 is the only signal that the dial was set too high, which is why it must run early enough for the frozen-encoder fallback to remain executable.

Gate: held-out generator family performance does not collapse. This is the validity check, not a tuning signal — it is consulted once, and its failure triggers the prepared fallback rather than a retune.

8.7 S5 — Reliability fitting

No gradient descent. With the detection model frozen, compute quality descriptors on the authentic null pool, form quality cells, and estimate per-cell μ and σ with shrinkage toward global authentic statistics for sparse cells. On the development pool, compute $d'(k,b)$ per cell and derive the availability table $a_k(x)$.

Audit cell occupancy before trusting the table: confirm that noise-degraded images occupy distinct cells rather than collapsing into clean ones, and that no cell is so sparse its statistics are noise.

Gate: predicted a_k tracks measured d' on the held-out transform family. Failure disqualifies availability weighting; fall back to passive quality features in the stacker.

8.8 S6 — Calibration and freeze

Fit temperature T and any operating thresholds on the calibration pool only, after every architectural and reliability decision is final. Derive the risk–coverage curve for the insufficient-evidence reporting overlay.

Gate: all weights, tables and thresholds frozen. The locked set is opened once, after this point, and no result from it may alter any of the above.

8.9 Failure paths, decided in advance

If	Then
S0 nuisance probe fires	Canonicalise and re-run. Do not proceed to training.
S2 prototypes collapse	Re-initialise from k-means with stronger diversity weight; reduce K.
S3 curve still rising at max K	Proceed with the largest feasible K, record undersizing, and raise the expected worst-authentic-subtype FPR in the report.
S4 worst-subtype loss rises	Stop. The reference is rejecting authentic long-tail content; reducing adaptation strength is the first response.
S4 held-out family collapses	Freeze the encoder, retrain heads only against the same memory. Weaker, but valid and shippable.
S5 availability fails its gate	Ship passive quality features in the stacker; report the negative result.

9 Objective

$$\mathcal{L} = \mathcal{L}_{\text{BCE-bal}} + \lambda_p \mathcal{L}_{\text{pAUC}} + \lambda_{\text{pair}} \mathcal{L}_{\text{pair}},$$

with balanced cross-entropy weighted by master, class, authentic subtype and generator, and a structured partial-AUC surrogate over the hardest negatives:

$$\mathcal{L}_{\text{pAUC}} = \frac{1}{|P||N_\alpha|} \sum_{i \in P} \sum_{j \in N_\alpha} \text{softplus}(m - z_i + z_j), \qquad \alpha = 0.05.$$

CVaR / group-DRO is not a default: it requires strong regularisation and early stopping on overparameterised networks, and with quality cells as an additional grouping axis several groups are small enough to chase label noise. Gradient conflict between terms is monitored by $C_{ij} = \nabla\mathcal{L}_i \cdot \nabla\mathcal{L}_j / (\|\nabla\mathcal{L}_i\| \|\nabla\mathcal{L}_j\|)$, with persistent negative C_{ij} and no held-out gain resolved by staging rather than reweighting.

10 Evaluation

10.1 Operating point and power

To estimate FPR at level α with relative standard error ε,

$$n_{\text{real}} \approx \frac{1-\alpha}{\alpha\,\varepsilon^2}.$$

At $\alpha = 0.01,\ \varepsilon = 0.25$: on the order of 1,600 authentic masters. At $\varepsilon = 0.10$: on the order of 9,900. At $\alpha = 0.05,\ \varepsilon = 0.25$: on the order of 300.

> **Consequence.** Select on normalised partial AUC over FPR $\in [0, 0.05]$ and TPR at 5% FPR. Report TPR at 1% FPR with Wilson or exact binomial intervals as a diagnostic. Do not tune choices against a statistic supported by a handful of tail observations.

10.2 Metrics and transform analysis

ROC-AUC and average precision; normalised pAUC over $[0, 0.05]$; TPR at 5% and 1% FPR with intervals; Brier and ECE; risk at coverage; worst-group values over predeclared generator, authentic-source, journey and **quality-cell** groups. Component deltas are reported with master-clustered paired bootstrap intervals, resampling lineages so clean and transformed views stay paired.

Signal survival $\rho(k,t) = d'(k,t)/(d'(k,\text{clean}) + \varepsilon)$ is reported **with class means separated**. A collapse driven by authentic scores rising toward the synthetic class is a false-positive mechanism and is far more damaging than an equal collapse driven by synthetic scores falling; the ratio alone conceals the difference.

Interaction is measured by $I(a,b) = d'(a \text{ after } b) - d'(a) - d'(b) + d'(\text{clean})$. If only one chain is tested it should be resize-then-JPEG: the realistic redistribution pipeline and, under H1 and H2, the composition most likely to be super-additively destructive.

11 Validity check and gates

11.1 The single measurement that validates the approach

Because the encoder is adapted, one question determines whether the system is worth submitting: **does it generalise to a generator family it never saw?**

Hold out an entire generator family from training — not random images, the whole family. Evaluate on it once, before the locked set is opened. This is not a component ablation; it is a validity check on the architectural commitment, and it costs a single evaluation pass.

Fallback, decided in advance. If held-out-family performance collapses, LoRA has overfit. The prepared fallback is to freeze the encoder and take whatever the memory yields frozen — a weaker model, but a real one, and known in time to ship. Schedule this check early enough that the fallback remains executable.

11.2 Remaining gates

Pre-register $\delta_{\min} = \max(1 \text{ normalised-pAUC point}, 2 \times \text{finalist seed SD})$ before any comparison. A single-seed sub-$\delta_{\min}$ difference is not evidence.

- **Protocol.** No lineage, duplicate or DDA-pair leakage; metadata-only nuisance AUC near chance both before and after training; resampler parity verified; patch-extractor behaviour at low resolution documented. Fix data before touching architecture.
- **Capacity.** The authentic-tail coverage curve has flattened at the chosen K. If it has not, record the undersizing and expect elevated worst-authentic-subtype FPR.
- **H1.** The lattice split appears in the measured $\rho(k,t)$ table. If blur damages local evidence as much as JPEG does, the mechanism differs from the one assumed here and both the aggregation argument and the resolution-based availability rule require revision.
- **Availability.** a_k-weighted fusion beats passive quality-as-features by $\geq \delta_{\min}$, *and* predicted a_k tracks measured d' on the held-out transform family. Failing the second disqualifies the component even if the first passes.
- **Optional forensic branch.** Admitted only on fused pAUC gain $\geq \delta_{\min}$ with a positive paired lower bound, worst-authentic-subtype FPR rising by at most one point, and latency within budget.

12 Assumptions and residual risks

1. **H1 is inherited, not measured** on this system's branches. The single-transform survival sweep must precede architectural commitment.
2. **Transfer of $d'(k,b)$ across transform families** is the load-bearing assumption of the reliability layer, and is more fragile under adaptation than it would be for a frozen model.
3. **LoRA is a bet against a published finding.** The robustness case for frozen probes is that they resist degradation *because* they cannot overfit fragile artifacts. Adaptation moves back toward the regime those results show failing under compression. Published reference-comparison robustness under adaptation is demonstrated at mild severity (JPEG $q \approx 90$, resize ≈ 0.9); the hard cases here are $q30$ and $0.25\times$, where no evidence exists either way. This is the principal risk in the design and it is taken knowingly.
4. **Resize $0.25\times$ may be irreducible.** Where evidence is erased, the correct behaviour is a calibrated low-confidence score, not a confident guess.

5. **Adversarial robustness is out of scope.** Diffusion purification, regeneration and adversarial perturbation defeat this class of detector and are not claimed against. Removing the intervention set reduces but does not eliminate the attack surface.

6. **Partial manipulation is out of scope.** The binary formulation assumes fully synthetic versus authentic origin.

7. $q(x)$ **appears on both sides.** It determines availability and may enter the stacker. Inspect the fitted quality weights and re-run the nuisance probe on the trained model to confirm quality has not become a class proxy.

13 Novelty boundary

The defensible contribution is the **replacement of per-image intervention response with per-cell measured survival** as the reliability signal, together with the noise-floor correction that makes quality cells well-posed under masking operators, evaluated with class-separated survival analysis under pre-registered gates.

Not claimed as invented: reference-comparison detection against a real-only memory bank; dual data alignment; frozen visual-foundation-model probing; partial-AUC optimisation; perturbation-response detection. Each is prior art and must be cited by name. The contribution is the reliability formulation and the tested integration, not the mechanisms.

The success criterion is not that every module survives. It is that the build ends with a precise statement of which hypothesis was tested, under which controls, and with what interval.